Heart Disease Prediction Using Machine Learning

Objective:

To develop a machine learning model that predicts the likelihood of heart disease based on patient data such as age, cholesterol, blood pressure, lifestyle factors, etc.

1. Libraries Used:

- pandas
- numpy
- matplotlib
- seaborn
- scikit-learn

2. Dataset Description:

The dataset contains various features about patient health and habits including:

- Age, Gender, Cholesterol, Blood Pressure, Heart Rate
- Smoking, Alcohol Intake, Exercise Hours
- Family History, Diabetes, Obesity, Stress Level
- Blood Sugar, Chest Pain Type, Exercise Induced Angina
- Target: Heart Disease (1 = Yes, 0 = No)

3. Preprocessing Steps:

- Handled missing values: numeric filled with mean, categorical with mode
- Converted categorical values using Label Encoding
- Standardized features using StandardScaler

4. Exploratory Data Analysis (EDA):

- Correlation heatmap created to find relationships between features
- Scatter plots visualized key relationships like Age vs Cholesterol

5. Model Training:

• Split data into train and test sets using train_test_split
• Used RandomForestClassifier to train model
• Evaluated with Accuracy, Classification Report, and Confusion Matrix

6. Prediction Function:

• A function was created to predict heart disease from new input data.
• It encodes and scales inputs before making a prediction using the trained model.

Why This Prediction is Important:

Heart disease is one of the leading causes of death globally. Early detection of heart disease can significantly improve treatment outcomes and even save lives. By using machine learning to analyze patient health data, we can:

• Identify high-risk individuals even before they show severe symptoms
• Help doctors prioritize patients who need urgent intervention
• Provide personalized health recommendations based on risk factors
• Reduce healthcare costs through preventive care

7. Summary:

• The model successfully predicts heart disease with high accuracy.
• Important factors: Age, Cholesterol, Blood Pressure, Lifestyle Habits
• The project shows how ML can help in early healthcare diagnosis.